

How Different Sound Levels Can Affect You - Noise Project

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Noise pollution can manifest itself in many ways like talking, transportation, or construction. These different ways relate to different ranges of sound levels. Sound levels are measured in decibels, which can range from 0 to over 120. Decibels are calculated using sound meters, like the one used in the “Map the Sound” button on the Noise Project app. Total silence equals 0 decibels (dB). If a sound level increases by 10 on the decibel scale, that means the sound is ten times more intense. But, what do these sound levels actually mean for you and your health? It is important to remember that sound and noise are two different things. Sound refers to what we’re hearing, while noise is more about how a sound makes us feel. Noise is an unwanted sound—a noise that makes you annoyed, stressed, distracted, or mad. Although noise is an unwanted sound, people can (or have to) become accustomed to it because they can’t control it—they often normalize noise to make life more bearable. Below is a list of different sound level ranges, what they sound like, and how they can affect your health.

0 dB to 30 dB This decibel range is very low risk. It is equivalent to sounds that are not very audible, including a ticking watch or normal breathing. It may be difficult to hear at this level if you experience hearing loss. This range is not associated with any negative health effects, and is actually correlated with times of relaxation and good sleep.

30 dB to 40 dB This decibel range is low risk. It is equivalent to sounds that are audible, but still quiet—think of soft whispering (3 feet away), rustling leaves, and mosquitos buzzing. It may also be difficult to hear at this level if you have hearing loss. This range is not associated with any specific health problems—but, anything exceeding this sound level can cause damage to your sleep patterns.

40 dB to 50 dB This decibel range is low risk. It is equivalent to sounds that are noticeable and can become disruptive depending on the environment—the humming of a fridge, a running stream, or other background noises. This range is associated with a lack of sleep, and prolonged exposure may also cause headaches and irritability. Chronic exposure to sound levels above this range can also lead to impaired cognitive function, especially in children. A study performed in 2008 found that noise led to primary school children developing a lack of motivation and reduced memory and reading comprehension. Children exposed to higher levels of noise had lower test scores. In general, the children in this study found it harder to focus because of the constant exposure to noise.

50 dB to 60 dB This risk level is considered to be normal—but just because it is normal doesn’t mean it can’t have negative impacts! It is equivalent to sounds that are easily noticeable and may compete with what you are trying to listen to and focus on, such as rain muffling, a conversation, or noise in a quiet office. This range is associated with an increased risk of cardiovascular disease. A study in 2019 found that road-traffic noise (starting at 50 dB) increased the risk of heart disease by 8% for every 10 dB. This is mainly because noise at this level and above can cause an increase in stress hormones, either through sleep deprivation, anger/

annoyance, or disrupting your performance or communication. Stress hormones increase blood pressure, blood sugar, and the constriction of blood vessels. These factors predispose people to cardiovascular disease and diabetes (hence the increased blood sugar). Cardiovascular disease can develop into coronary heart disease, heart failure, arrhythmia (irregular heartbeats), and strokes.

60 dB to 70 dB This risk level is also considered to be normal. It is equivalent to sounds that are easy to hear and may become annoying if you can't control them, including laughter, a loud conversation, a hair dryer, or a vacuum. According to the American Speech-Language-Hearing Association, you can listen to this sound range for as long as you want without developing hearing loss. Of course, listening to this sound range for prolonged periods of time (for more than eight hours) can still lead to the more indirect health effects listed in the previous sections.

70 dB to 80 dB This risk level is considered to be high, meaning that it takes less time in this sound range to experience damage to your health. It is equivalent to sounds that feel too loud and may overpower what you are listening to, including cars, a dishwasher, a washing machine, and a garbage disposal. You may feel very annoyed by the noise. Prolonged exposure to noise at this range may begin to make you feel uncomfortable. This range is associated with Noise Induced Hearing Loss (NIHL). NIHL tends to become a risk at 70 dB and above. The ear is sensitive and its internal structures, like hair cells (which protect the ear from vibrations), can become damaged at this sound level (visit NIH for more info). NIHL makes it more difficult to hear and communicate, and in severe cases may require the use of hearing aids. One in four American adults shows signs of hearing loss due to noise pollution. Some people are more susceptible to developing NIHL, such as the elderly, so they should especially be aware and careful when in this sound range.

80 dB to 90 dB This risk level is also considered to be high and harmful to those who experience it! It is equivalent to sounds that can be startling and disruptive, including city traffic and a lawn mower. This range is associated with tinnitus. Tinnitus means you hear a sound that doesn't match the external source, such as a high-pitched ringing or buzzing sound. It can last anywhere from a few minutes to days and often causes distress. It is recommended that you only be exposed to this sound level for a maximum of 8 hours, although damage to hearing is possible after 2 hours. This sound level and above can also have adverse effects on pregnant women. Pregnant women exposed to this noise experience more stress and have been found to have an increased risk of having an underweight fetus and newborn. The mother was also at a higher risk of developing high blood pressure (gestational hypertension) and their infant was also at a higher risk of developing congenital malformations, including congenital heart disease. Noise directly affects the fetus, as sound is easily transmitted to the fetus, as well as indirectly. The increased stress hormones of the mother can pass into the placenta and release fetal stress hormones leading to these effects.

90 dB to 100 dB This risk level is considered to be dangerous, meaning that it can have a negative impact on your hearing in just under an hour. It is equivalent to sounds including a subway train, a car horn (at 16 feet), sporting events, a farm tractor, and a garbage truck. This range is associated with behavioral changes. The loud noise can cause someone to have an aggressive and violent reaction, which if occurring often (making it hard to cope), can lead to personality changes and mental health issues. When this sound range (and above) is combined with strong vibrations it can cause muscle spasms and break a bone. Additionally, many hospitals tend to be in this sound range and may cause patients' healing process to be slower.

100-150 dB This risk level is considered to be extremely dangerous and painful, meaning that it can have a negative impact on your hearing in just fifteen minutes! It is equivalent to sounds including a plane (flying 100 feet above you), a snowmobile, jackhammer, orchestra, power saw, boom box, thunderclap, stereo, concert, shotgun firing, and fireworks. This range is associated with ruptured eardrums, which is very painful. The eardrum is thin and separates the inner and outer ear. Ruptured eardrums can cause dizziness, ear drainage, earaches, and severe hearing loss. The sound of mining is included in this sound level range. Mining and hearing impairment are strongly related. It has been shown that 70% of male miners will experience hearing impairment by the age of 60. The risk and severity of hearing loss increases greatly in this sound range. Severe hearing loss may cause someone to be five times more likely to develop dementia than someone without hearing loss. So How

Can You Prevent This? There are a few simple ways you can prevent experiencing the negative effect of noise pollution. Sometimes, it may be out of your control, due to where you live and the resources available to you, but there are a number of solutions that have proved to help. One solution is going to your nearest noise refuge. Noise refuges are spaces that shield us from the chaotic noises in our environment. Common noise refuges include green spaces, like parks, or even libraries. Another solution includes getting personal protective equipment (PPE), like ear plugs. Even though you may not have access to PPE or other resources to protect yourself, you should try to reduce the effects of noise pollution by any means possible. If this means using toilet paper to stuff your ears instead of ear plugs, do just that! Improvise! Be resourceful! A very impactful solution would be to get involved in noise pollution policy advocacy, which is important because inequity and someone's socioeconomic background plays a huge role in how much noise they're exposed to. None of these solutions can be done without recognizing how noise pollution affects you first. The Noise Project app can help you do this by simply recording the sound levels in your surroundings. Then, you'll be able to better understand how to respond to noise pollution in your environment! You can also help your community and others to protect themselves from noise pollution by sharing noise refuges you have found. This is made simple by the Noise Project app in which you can use the "Map the Sound" button to record the sound level and then suggest your location as a noise refuge. Once you do that, anyone can see and visit the noise refuge on the Map in the app. You can also map locations with dangerous sound levels so others can avoid them. Your simple push of a button might save someone from experiencing the damaging effects of noise pollution!